

Midterm Exam Solution

General Departmental Principles

Partial credit is awarded for any idea or partial solution from which, with suitable completion of the reasoning in the script, a correct full solution could be obtained. If a student starts several substantially different approaches to a problem, at most one can be credited. If all presented approaches or fragments are correct or could be completed to be correct, the attempt that yields the most partial credit should be evaluated. However, if among several attempts there is at least one correct and at least one with a (substantial) error, and it is not clear from the script which one the student considered correct, then the lower-scoring attempt should be graded (even if this score is 0). Partial scores shown in the guide may be further subdivided if necessary. A good solution that differs from what is written in the guide naturally earns full marks.

In case of an arithmetic error, deduct 1 point per error within a problem. Exception: if the error significantly simplifies or changes the structure of the problem. In such cases, no points are awarded for those subparts that, due to the error, did not even arise.

1. We know about events A , B , and C that at least one of A or B always occurs, A and C are independent, and $\mathbb{P}(A) = \mathbb{P}(B) = 0.6$ and $\mathbb{P}(C) = 0.1$. Answer (with justification!) the following questions.
 - (a) What is the probability that A and C both occur (simultaneously)?
 - (b) What is the probability that A and B both occur (simultaneously)?
 - (c) What is $\mathbb{P}(B|A)$?
 - (d) What is the probability that neither A nor C occurs?

Solution.

(a) (4 points)

(1 pt) The desired probability is $\mathbb{P}(A \cap C)$ (it is not necessary to write “the desired probability”; it should just be clear what is being computed).

(1 pt) $= \mathbb{P}(A)\mathbb{P}(C)$

(1 pt) $= 0.06$ (a common fraction is also acceptable),

(1 pt) because they are independent.

(b) (7 points)

(1 pt) The desired probability is $\mathbb{P}(A \cap B)$.

(1 pt) By the Poincaré formula (the inclusion–exclusion formula is also OK, but it must be named)

(2 pts) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$,

(1 pt) rearranging gives $\mathbb{P}(A \cap B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cup B)$. (Can be combined with the previous line, but only if stated correctly; otherwise no partial credit.)

(1 pt) Since at least one of A or B always occurs, $\mathbb{P}(A \cup B) = 1$.

(1 pt) Hence $\mathbb{P}(A \cap B) = 0.6 + 0.6 - 1 = 0.2$ (OK without further detail; $\frac{1}{5}$ is also acceptable).

(c) (3 points)

(1 pt) (By definition—no need to write this) $\mathbb{P}(B|A) = \mathbb{P}(A \cap B)/\mathbb{P}(A)$

(1 pt) $= 0.2/0.6$

(1 pt) $= \frac{1}{3}$ (a decimal is also OK).

(d) (6 points)

(1 pt) The desired probability is $\mathbb{P}(\overline{A} \cap \overline{C})$.

Solution 1.

(2 pts) Since A and C are independent, $\bar{A}$ and $\bar{C}$ are also independent.

(1 pt) Thus $\mathbb{P}(\bar{A} \cap \bar{C}) = \mathbb{P}(\bar{A})\mathbb{P}(\bar{C})$

(1 pt) $= (1 - \mathbb{P}(A))(1 - \mathbb{P}(C))$

(1 pt) $= 0.4 \cdot 0.9 = 0.36$ (OK without detailed computations; a common fraction is also acceptable).

Solution 2.

(1 pt) $\mathbb{P}(\bar{A} \cap \bar{C}) = \mathbb{P}(\overline{A \cup C})$.

(1 pt) $= 1 - \mathbb{P}(A \cup C)$.

(0 pt) Again by the Poincaré formula

(1 pt) $\mathbb{P}(A \cup C) = \mathbb{P}(A) + \mathbb{P}(C) - \mathbb{P}(A \cap C)$

(1 pt) $= 0.6 + 0.1 - 0.06 = 0.64$,

(1 pt) hence the desired probability is 0.36.

2. At the reception of Building I there are 2 blue, 3 green, 1 black, and 2 red markers. Before a probability class we randomly (uniformly) select one of the markers to write on the board. We know that a blue marker runs out by the end of class with probability $\frac{2}{5}$, a green with probability $\frac{1}{4}$, a black with probability $\frac{1}{5}$, and a red with probability $\frac{1}{10}$.

- What is the probability that we choose a green marker and it runs out by the end of class?
- What is the probability that the chosen marker (regardless of color) runs out by the end of class?
- Given that the chosen marker ran out by the end of class, what is the probability that it was red?

Solution.

Let K, Z, F, P denote the events that we select a blue, green, black, or red marker, respectively, and let B denote the event that the selected marker runs out. (Of course, naming these is not necessary if the corresponding theorems are used correctly.)

(a) (5 points)

(1 pt) The desired probability is $\mathbb{P}(Z \cap B)$

(1 pt) $= \mathbb{P}(Z)\mathbb{P}(B|Z)$ (if this is present, the previous point can also be awarded)

(1+1 pts) $= \frac{3}{8} \cdot \frac{1}{4}$ (no further justification needed; $\mathbb{P}(Z)$ equals this since 3 of 8 markers are green and the space is classical)

(0 pt) $= \frac{3}{32} (= 0.09375)$,

(1 pt) where in the first equality we used the product rule (no need to name it if it is derived from the definition of conditional probability).

(b) (8 points)

(1 pt) The desired probability is $\mathbb{P}(B)$.

(1 pt) By the Law of Total Probability (LOTP)

(1 pt) $\mathbb{P}(B) = \mathbb{P}(K)\mathbb{P}(B|K) + \mathbb{P}(Z)\mathbb{P}(B|Z) + \mathbb{P}(F)\mathbb{P}(B|F) + \mathbb{P}(P)\mathbb{P}(B|P)$

(2 pts) $= \frac{2}{8} \cdot \frac{2}{5} + \frac{3}{8} \cdot \frac{1}{4} + \frac{1}{8} \cdot \frac{1}{5} + \frac{2}{8} \cdot \frac{1}{10}$ (as in (a); no need to justify)

(1 pt) $= \frac{16+15+4+4}{160} = \frac{39}{160} (= 0.24375)$. (Also acceptable without further detail if the source of the expression is clear.)

(1 pt) We can use LOTP because K, Z, F, P form a complete system of events,

(1 pt) being pairwise disjoint and with union equal to the entire sample space (if this is stated, the previous point is also awarded; no further explanation needed as it is evident

we draw exactly one marker and always draw some color).

Here too, deriving LOTP (even just this specific form) from the definition of conditional probability suffices, without naming the theorem.

(c) (7 points)

(1 pt) The desired probability is $\mathbb{P}(P|B)$.

(1 pt) By Bayes' theorem (Bayes' formula, or anything explicitly naming Bayes; the "simple Bayes" is not enough, but full correct formulas earn credit)

(2 pts)

$$\mathbb{P}(P|B) = \frac{\mathbb{P}(B|P)\mathbb{P}(P)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B|P)\mathbb{P}(P)}{\mathbb{P}(K)\mathbb{P}(B|K) + \mathbb{P}(Z)\mathbb{P}(B|Z) + \mathbb{P}(F)\mathbb{P}(B|F) + \mathbb{P}(P)\mathbb{P}(B|P)}$$

(either form suffices. In the first form, referencing the product rule and that $\mathbb{P}(B)$ was already computed is also OK. If the product rule was referenced in (a), it need not be cited again.)

$$(1+1 \text{ pts}) = \frac{\frac{2}{8} \cdot \frac{1}{10}}{\frac{39}{160}}$$

$$(1 \text{ pt}) = \frac{4}{39} (\approx 0.1026).$$

3. On a certain tram line, each day an operational failure occurs with probability $\frac{1}{10}$, independently from day to day.

(a) Over 20 days we observe each day whether a failure occurred. Let X be the number of days (out of 20) on which a failure occurred. Give the expectation and standard deviation of X , and compute $\mathbb{P}(X > 2)$.

(b) Starting from a given day—call this Day 1—we observe daily whether a failure occurs, until the first failure. What is the probability that this first failure occurs exactly on Day 3? And given that there was no failure on any of the first three days, what is the probability that there is no failure on any of the first five days?

Solution.

(a) (10 points)

(1 pt) X has a binomial distribution

(1+1 pts) with parameters $n = 20$ and $p = \frac{1}{10}$. (It suffices to write $X \sim \text{Bin}(20, \frac{1}{10})$. No justification required, but the usual reasoning applies: n independent trials with success probability p , where "success" is a failure, and X counts successes. Notation n, p is not mandatory if parameter use is otherwise clear.)

(1 pt) Hence $\mathbb{E}(X) = np = 2$.

(1 pt) And $\mathbb{D}(X) = \sqrt{\mathbb{D}^2(X)}$

(1 pt) $= \sqrt{np(1-p)}$ (having this, even with n, p substituted, earns the previous point as well)

(1 pt) $= \sqrt{1.8} \approx 1.3416$.

(1 pt) $\mathbb{P}(X > 2) = 1 - \mathbb{P}(X = 0) - \mathbb{P}(X = 1)$

(1 pt) $= 1 - (1 - 0.1)^{20} - 20 \cdot 0.1 \cdot (1 - 0.1)^{19}$ (also OK with n, p ; here we used $\binom{20}{0} = 1$, $\binom{20}{1} = 20$)

(1 pt) ≈ 0.6083 .

(b) (10 points)

(1 pt) Let Y be the index of the first day on which a failure occurs. Then Y has a

geometric distribution

(1 pt) with parameter $p = \frac{1}{10}$ (again, $Y \sim \text{Geo}(\frac{1}{10})$ suffices).

(1 pt) The first desired probability is $\mathbb{P}(Y = 3)$

(1 pt) $= (1 - 0.1)^{3-1} 0.1$ (using p is also OK)

(1 pt) $= 0.081$.

(1 pt) The other desired (conditional) probability is $\mathbb{P}(Y > 5 | Y > 3)$,

(1 pt) which by the memoryless property of the geometric distribution

(1 pt) equals $\mathbb{P}(Y > 2)$,

(1 pt) and $\mathbb{P}(Y > 2) = 1 - \mathbb{P}(Y = 1) - \mathbb{P}(Y = 2) = 1 - 0.1 - 0.9 \cdot 0.1$

(1 pt) $= 0.81$.

(For example, writing $\mathbb{P}(Y \geq 4 | Y \geq 2)$ is also acceptable, though using memorylessness correctly may be a bit harder in that form.)

4. We place into one hat a slip with the number 3 and a slip with the number 4. Into another hat we place a slip with the number 2 and a slip with the number 3. We draw one slip at random from each hat. Define random variables X and Y as follows: let $X = 2$ if the number drawn from the first hat is 4, and $X = 1$ otherwise; let Y be the number of hats from which the number 3 was drawn. (For example, if we draw 3 from the first hat and 2 from the second, then $X = 1$ and $Y = 1$.)
- Determine the joint distribution of X and Y (i.e., $\mathbb{P}(X = k, Y = \ell)$ for all k, ℓ with positive probability) in tabular form.
 - Determine the marginal distributions of X and Y .
 - Are X and Y independent?
 - Determine the expectation of X^3 (note: third power!).

Solution.

$Y \backslash X$	1	2	p_Y
0	0	$\frac{1}{4}$	$\frac{1}{4}$
1	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{2}$
2	$\frac{1}{4}$	0	$\frac{1}{4}$
p_X	$\frac{1}{2}$	$\frac{1}{2}$	(1)

Scoring:

(a) (6 points)

(6 pts) Perfect joint distribution table.

If not perfect: award 1 point for each correct nonzero entry.

If any negative probability or a probability greater than 1 appears in the table, then part (a) immediately gets 0 points!

If the probabilities in the table do not sum to 1, then the total for (a) is at most 3 points.

(b) (5 points)

(3 pts) The pmf values of Y are obtained by summing the appropriate rows (no justification needed)—it's enough to apply this, not merely state it. If only stated but not carried out, award 1 point.

(2 pts) The pmf values of X are obtained by summing the appropriate columns.

If X and Y are swapped, then (a) is worth at most 4 and (b) at most 2 points. It is

allowed to draw the transposed table, but then the marginals must be computed accordingly.

Do not deduct again here for errors already made in (a), but negative probabilities or values greater than 1 still receive no credit; and if a marginal does not sum to 1, that marginal is worth at most 1 point.

(c) (4 points)

A bare statement “not independent” without justification earns no points.

(1 pt) Not independent, because... (some reason—even if not fully correct, as long as it reflects genuine reasoning).

(1 pt*) For independence we would need $\mathbb{P}(X = k, Y = \ell) = \mathbb{P}(X = k)\mathbb{P}(Y = \ell)$ for every $k \in \text{Ran}(X)$ and $\ell \in \text{Ran}(Y)$.

(1 pt) Exhibit a pair (k, ℓ) where $\mathbb{P}(X = k, Y = \ell) \neq \mathbb{P}(X = k)\mathbb{P}(Y = \ell)$ (can be combined with the previous step).

(1 pt) Verify these are indeed unequal, e.g., by writing the specific values of $\mathbb{P}(X = k, Y = \ell)$ and $\mathbb{P}(X = k)\mathbb{P}(Y = \ell)$ (no need to over-explain). In particular, it suffices if the joint probability is 0 while the product of marginals is nonzero.

**Note:* Since we only need to show *non*-independence, a single counterexample with the explanation that the intersection probability is not the product of the marginals suffices for full credit on (c), even without explicitly stating that the equality must hold for all k, ℓ in the independent case.

(d) (5 points)

(0 pt) By the formula for the expectation of a transform,

(3 pts) $\mathbb{E}(X^3) = 1^3 \mathbb{P}(X = 1) + 2^3 \mathbb{P}(X = 2)$.

(No need to start from a more general formula. If only a more general statement is given—for example, involving $g(X)$ or $\text{Ran}(X)$ —but not specialized correctly, award at most 2 points. No credit if the probabilities are cubed instead of the values.)

(1 pt) $= \frac{1}{2} + 8 \cdot \frac{1}{2}$

(1 pt) $= \frac{9}{2}$ (= 4.5).

5. A continuous random variable X has density

$$f_X(x) = \begin{cases} ax, & \text{if } 1 < x < 2, \\ 0, & \text{otherwise,} \end{cases}$$

where a is a suitably chosen real constant.

(a) Determine the value of a .

(b) Determine the distribution function of X .

(c) Determine the expectation of X .

Solution.

(a) (7 points)

(1 pt) For f_X to be a density it must be nonnegative (so $a \geq 0$ —this will follow from the solution; it is enough to mention this once).

(1 pt) We also need $1 = \int_{-\infty}^{\infty} f_X(x) dx$

(2 pts) $= \int_1^2 ax dx$ (can be combined with the previous line)

(1 pt) $= [ax^2/2]_1^2$

$$(1 \text{ pt}) = a(2^2/2 - 1^2/2) = \frac{3}{2}a,$$

$$(1 \text{ pt}) \text{ hence } a = \frac{2}{3}.$$

(b) (7 points)

(1 pt) If $x \leq 1$, then $F_X(x) = 0$ (since $F_X(x) = \mathbb{P}(X < x)$ and $\text{Ran}(X) = (1, 2)$).

(1 pt) If $x \geq 2$, then $F_X(x) = 1$.

(1 pt) If $1 < x < 2$, then $F_X(x) = \int_{-\infty}^x f_X(t) dt$

(2 pts) $= \int_1^x \frac{2}{3}t dt$ (can be combined with the previous line)

(1 pt) $= [\frac{1}{3}t^2]_1^x$

(1 pt) $= \frac{1}{3}x^2 - \frac{1}{3}.$

(c) (6 points)

(1 pt) $\mathbb{E}(X) = \int_{-\infty}^{\infty} x f_X(x) dx$

(2 pts) $= \int_1^2 \frac{2}{3}x^2 dx$

(1 pt) $= [\frac{2}{9}x^3]_1^2$

(2 pts) $= \frac{2}{9} \cdot 8 - \frac{2}{9} \cdot 1 = \frac{14}{9}.$

6. * On the open interval $(0, 2)$ we choose at random (with a uniform distribution) a real number y . In the plane $\mathbb{R}^2$, consider the right triangle with vertices $(0, 0)$, $(1, 0)$, and $(0, y)$. Let X denote the length of the hypotenuse of this triangle (i.e., the distance between $(1, 0)$ and $(0, y)$).

(a) What is the probability that X is less than 2?

(b) Determine the distribution function of X .

(c) Determine the density of X .

(d) Determine the expectation of $Z := \frac{2\sqrt{X^2 - 1}}{X}$. (It is not required to determine the distribution of Z .)

Solution.

(a) (7 points)

(1 pt) By the Pythagorean theorem, $X^2 = 1^2 + y^2$ (no need to explicitly cite the theorem).

(1 pt) Hence $\text{Ran}(X) = (1, \sqrt{5})$ (it suffices if this becomes clear later).

(1 pt) The desired probability is $\mathbb{P}(X < 2)$

(1 pt) $= \mathbb{P}(1 + y^2 < 2^2) = \mathbb{P}(y < \sqrt{3})$ (no absolute value is needed since $y > 0$ —or, once the range is stated, no separate justification is needed).

(1 pt) $= \frac{\text{favorable length}}{\text{total length}}$ (some reference to the 1D geometric probability model or to the uniformity of y ; a good figure also counts as justification).

(1+1 pts) $= \frac{\sqrt{3}}{2} \approx 0.8660$ (1 pt for the favorable and 1 pt for the total length; giving either the formula or the numeric value is sufficient).

(b) (5 points)

(1 pt) For any real x , $F_X(x) = \mathbb{P}(X < x)$.

(1 pt) Thus $F_X(x) = 0$ if $x \leq 1$,

(1 pt) and $F_X(x) = 1$ if $x \geq \sqrt{5}$.

(2 pts) For $1 < x < \sqrt{5}$, the same computation as in part (a) with x in place of 2 gives

$\mathbb{P}(X < x) = \mathbb{P}(1 + y^2 < x^2) = \mathbb{P}(y < \sqrt{x^2 - 1}) = \frac{\sqrt{x^2 - 1}}{2}$. (As justification, it suffices to carry out the computation—no need to re-explain geometric probability—or to state that this is analogous to (a) with x in place of 2. With no justification at all, award at most 1 point.)

In parts (a)–(b), if the formal solution is missing but there is a good, informative figure showing what X is and how its length is computed via y , award 1 point. If the figure also clearly indicates the event $\{X < 2\}$ or $\{X < x\}$, award 1 additional point.

(c) (4 points)

(1 pt) For $1 < x < \sqrt{5}$, $f_X(x) = F'_X(x)$

(2 pts) $= \frac{2x}{2 \cdot 2\sqrt{x^2 - 1}}$ ($= \frac{x}{2\sqrt{x^2 - 1}}$). If there is only a small slip—e.g., a missing factor 2 or 1/2—award 1 of the 2 points. If the chain rule factor is omitted, no credit for these 2 points.)

(1 pt) $f_X(x) = 0$ if $x \leq 1$ or $x \geq \sqrt{5}$ (no separate justification needed. For $x < 1$ or $x > \sqrt{5}$, we also have $f_X(x) = F'_X(x)$).

(d) (4 points)

(0 pt) By the formula for the expectation of a transform,

(1 pt) $\mathbb{E}(Z) (= \mathbb{E}(\frac{2\sqrt{X^2-1}}{X})) = \int_{-\infty}^{\infty} \frac{2\sqrt{x^2-1}}{x} f_X(x) dx$

(1 pt) $= \int_1^{\sqrt{5}} \frac{2\sqrt{x^2-1}}{x} \cdot \frac{x}{2\sqrt{x^2-1}} dx$ (can be combined with the previous line)

(1 pt) $= \int_1^{\sqrt{5}} 1 dx$

(1 pt) $= [x]_1^{\sqrt{5}} = \sqrt{5} - 1 \approx 1.236$ (it is fine not to highlight the antiderivative separately, and either a formula or a numeric value suffices).